

GROUP 1: ADMINISTRATION, SECURITY AND MULTITENANCY

❖ Scope

- Role-Based Access Control (RBAC) with support for multiple organizations, hierarchies, and tenants (e.g., ministry → agency → facility)
- Single Sign-On (SSO) integration with government identity providers where available
- Full audit trails for changes to master data and transactions
- Configurable master data: asset categories, locations, cost centres, user roles, workflows, and approval levels
- Support for multitenancy: separate data, users, and configurations per tenant while maintaining centralized management

❖ Key deliverables

- RBAC system with hierarchical organization and multitenancy support
- SSO integration setup and configuration
- Audit trail logs and reporting interface
- Master data configuration tools (categories, locations, cost centres, roles, workflows)
- User management dashboard and admin controls
- Tenant management interface for onboarding, configuration, and data isolation
- Security and access management documentation

❖ Dependency Matrix:

- Depends on: **Nobody**,
- Who depends on them: **Each group for security(auth & authorization)**

GROUP 2: ASSET & PROPERTY REGISTER MODULE

❖ Scope

- Maintain a centralized asset register with unique IDs for all assets
- Support multiple asset classes: land, buildings, floors, rooms, infrastructure, plant & equipment, vehicles, ICT assets, furniture, specialised equipment

- Record key asset attributes: ownership, location (GIS), land title, use type, condition, acquisition date/method, cost, current & residual value, useful life, funding source, responsible unit
- Support parent–child asset relationships (e.g., building → floor → room → installed equipment)
- Track asset status: planned, active, under construction, in use, under maintenance, decommissioned, disposed
- Manage supporting documents: titles, plans, warranties, permits, photos with secure storage & metadata tagging

❖ **Key deliverables**

- Centralized, searchable asset register database
- Asset classification and hierarchy structure
- Comprehensive attribute capture per asset
- Asset status tracking dashboard
- Secure document management system linked to assets
- Metadata tagging and document retrieval functionality

❖ **Dependency Matrix:**

- Depends on: **Group 1** (security & auth),
- Who depends on them: **Group 3, 4, 5, 6, 7, 8, 9 (Both need assets to work with)**

GROUP 3: LIFECYCLE PLANNING & CAPITAL PROJECTS MODULE

❖ **Scope**

- Support long-term asset lifecycle planning (renewal, replacement, refurbishment, expansion)
- Record asset condition assessments and remaining useful life (RUL)
- Prioritise interventions based on risk and criticality
- Manage capital improvement projects (scope, budget, timelines, responsible parties)
- Link capital projects to affected assets
- Provide scenario analysis for CAPEX decisions (repair vs replace, NPV, lifecycle cost)

❖ **Key deliverables**

- Asset lifecycle management dashboard
- Condition assessment & RUL records per asset
- Intervention prioritisation reports
- Capital improvement project management tool
- CAPEX scenario analysis reports
- Linked asset-project database

❖ **Dependency Matrix:**

- Depends on: **Group 1** (security & auth), **Group 2** (buildings/assets locations).
- Who depends on them: **Group 5 (Financials)**, **Group 10 (Analytics)**

GROUP 4: MAINTENANCE AND OPERATIONS (CMMS) MODULE

❖ **Scope**

- Register maintainable assets with preventive & predictive maintenance schedules
- Generate, assign & track work orders (corrective, preventive, predictive)
- Maintain full maintenance history per asset (faults, parts, downtime, costs)
- Manage service contracts, SLAs & contractor performance
- Automated reminders & escalation for overdue tasks
- Mobile access for technicians - Mobile app with offline feature (jobs, status, photos, notes)

❖ **Key deliverables**

- Centralised register of maintainable assets with PM schedules
- Work order management system (corrective, preventive, predictive)
- Asset-level maintenance history and cost tracking
- Contract & SLA performance tracking dashboard
- Automated alerts, reminders & escalation engine
- Mobile-enabled technician app
- Photo & note capture linked to work orders

❖ **Dependency Matrix:**

- **Group 1 (security & auth)**, **Group 9 (Inventory)**, **Group 2 (buildings/assets locations)**.

- Who depends on them: **Group 10 (Analytics), Group 8 (Compliance reporting), Group 3 (Lifecycle data)**

GROUP 5: FINANCIAL AND VALUATION MODULE

❖ Scope

- Multiple valuation methods (cost, market, replacement)
- Track acquisition cost, improvements, impairments, revaluations and disposal proceeds.
- Automated depreciation calculations (IPSAS/IFRS compliant)
- Provide asset registers and schedules for integration with accounting/ERP systems.
- Lease management for landlord/tenant scenarios
- Budget planning and tracking (CAPEX/OPEX)
- Financial reporting and schedules
- Manage tenant/occupant data for government and private occupants.
- Track rent invoicing, payment status and arrears (with integration to financial systems where available).
- Alerts for key dates (renewal, rent review, expiry, break clauses).

❖ Key deliverables

- Depreciation and calculation engine
- Lease Management System
- Budget planning module
- Financial report generator

❖ Dependency Matrix:

- Depends on: **Group 1 (security & auth), Group 3 (Life cycle plans), Group 2 (buildings/assets locations).**
- Who depends on them: **Group 10 (Analytics), Group 8 (Compliance reporting)**

GROUP 6: LAND MANAGEMENT AND GIS INTEGRATION

❖ Scope

- Capture cadastral information: plot numbers, title numbers, tenure type, encumbrances, easements, land use
- Integrate with GIS for visualization of land and property assets, including layers for ownership, use, zoning, and services
- Support workflows for land acquisition, consolidation, subdivision, and wayleave management
- Record valuations and compensation for compulsory acquisition where relevant
- Maintain building and site layouts (blocks, floors, spaces), including space type, area, occupancy capacity, and usage
- Support space allocation to departments, projects, or tenants and track occupancy (staff/m², utilisation rates)
- Provide visual space plans (2D/3D drawings or integrated BIM references where available)
- Manage room bookings and shared facilities (meeting rooms, labs, auditoriums)
- Generate utilisation and occupancy reports for optimisation and planning

❖ **Key deliverables**

- Centralized cadastral and property database
- GIS-integrated visualization of land and assets with multiple layers
- Land management workflow engine (acquisition, consolidation, subdivision, wayleave)
- Asset valuation and compensation records
- Building/site layout repository with space details
- Space allocation and occupancy tracking dashboard
- Visual space plans (2D/3D/BIM) linked to assets
- Room booking and shared facilities management interface
- Reports on occupancy, utilisation, and space optimisation

❖ **Dependency Matrix:**

- Depends on: **Group 1** (Authentication), **Group 2** (buildings/assets locations).
- Who depends on them: **Group 4 (maintenance locations)**, **Group 7 (energy monitoring)**, **Group 9 (inventory locations)**

GROUP 7: ENERGY, SUSTAINABILITY AND SMART MONITORING MODULE

❖ Scope

- Capture utility consumption at building/asset level (electricity, water, gas, fuel)
- Compute sustainability KPIs (kWh/m², water/m², CO₂e emissions)
- Integrate with smart meters, BMS & IoT sensors for real-time monitoring
- Monitor temperature, occupancy, energy use & indoor air quality
- Sustainability & ESG performance dashboards
- Carbon footprint tracking for facilities and assets
- Track energy efficiency projects (LED, solar PV, HVAC upgrades)
- Evaluate savings and performance of sustainability initiatives

❖ Key deliverables

- Energy data management system
- IoT data ingestion pipeline
- Sustainability dashboard
- Carbon footprint calculator
- Utility bill tracking system

❖ Dependency Matrix:

- Depends on: **Group 1** (security), **Group 2** (buildings/assets), **Group 6** (spatial data)
- Who depends on them: **Group 10** (analytics), **Group 8** (environmental compliance)

GROUP 8: COMPLIANCE, RISK AND SAFETY MANAGEMENT MODULE

❖ Scope

- Statutory inspection & compliance tracking for assets (fire, lifts, electrical, boilers, pressure vessels, environmental permits)
- Asset-based risk registers (safety, environmental, operational risks)
- Risk mitigation and control action tracking per asset/facility
- Incident & near-miss reporting linked to assets
- Corrective and preventive action (CAPA) tracking
- Document management for asset certifications, licences & inspection reports

❖ **Key deliverables**

- Compliance tracking system
- Risk assessment module
- Incident management system
- Inspection scheduling engine
- Certification management

❖ **Dependency Matrix:**

- Depends on **Group 1(authentication), Group 2(Assets), Group 4 (maintenance history)**
- Who depend on them: **Group 3 (risk data for planning), Group 7 (Energy use and sustainability) Group 10 (compliance reporting)**

GROUP 9: INVENTORY, STORE AND ASSETS IN TRANSIT (MODULE)

❖ **Scope**

- Spare parts and consumables inventory management
- Barcode/RFID integration for asset tracking
- Purchase order management and approval workflows
- Supplier and vendor management
- Stock level monitoring
- Track tools and movable equipment issued to staff or contractors.

❖ **Key deliverables**

- Inventory management system
- Barcode/RFID scanning module
- Purchase order workflow
- Supplier management portal

❖ **Dependency Matrix:**

- Depends on **Group 1(authentication), Group 2(Assets), Group 4 (maintenance needs)**
- Who depend on them: **Group 4 (spare parts for maintenance), Group 5 (Procurement costs), Group 10 (Inventory analytics)**

GROUP 10: ANALYTICS, DASHBOARD AND REPORTING MODULE

❖ Scope

- Role-based configurable dashboards (Executives, Facilities, Finance, Technical)
- KPIs: asset value, utilisation, maintenance backlog, downtime, compliance
- Standard reports (asset register, condition, valuation, lease, energy, risk)
- Report export (PDF, Excel, CSV)
- Scheduled email report distribution
- Drill down and data filtering
- Ad-hoc querying & analytics ((e.g., assets by age, region, condition, cost centre)
- Centralized, real time reporting data source

❖ Key deliverables

- Report generation engine
- KPI calculation
- Data visualization components
- Predictive analytics models

❖ Dependency Matrix:

- Depends on **all groups** for data sources
- Who depend on them: **All users** (executives, managers etc)